

Javon Darby
Prof. Anna Devarakonda
ITAI 2372
26 Feb 2026

The AI Revolution: Smart Cities & Transportation

1. Smart Cities & Transportation (Core Idea)

- **Smart cities** use **AI, IoT, sensors, and real-time data** to improve how cities operate.
- Transportation is becoming:
 - **Connected** (vehicles + infrastructure communicate)
 - **Predictive** (AI predicts traffic and accidents)
 - **Automated** (self-driving systems and smart signals)

Key Technologies

- Artificial Intelligence (AI)
- Machine Learning (ML)
- IoT sensors
- 5G networks
- Computer Vision
- Edge computing

Main Benefits

- Reduced traffic congestion
- Faster emergency response
- Lower emissions
- Safer roads

2. Evolution of Transportation Systems

Past

- Manual traffic lights
- Limited data
- Reactive maintenance (fix problems after they happen)

Now

- AI-powered traffic signals
- Real-time data tracking
- Predictive maintenance (fix before failure)

Example

- AI can predict traffic **30–60 minutes ahead**.

3. Connected Vehicle Technology

V2X = Vehicle-to-Everything communication

- **V2V** – Cars communicate with other cars (collision prevention)
- **V2I** – Cars communicate with traffic lights and roads
- **V2P** – Cars detect pedestrians

Impact

- Fewer accidents
- Better fuel efficiency
- Faster travel times

4. Smart Public Transportation

AI improves:

- Route optimization
- Wait times
- Maintenance prediction
- Crowd management

Examples

- AI predicts when buses will be full

- Real-time updates for passengers

5. Smart Parking Systems

- Sensors detect open spaces
- AI predicts parking availability
- Automatic payment systems

Benefits

- Less traffic from drivers searching for parking
- Reduced emissions

AI in Media & Content Creation

6. AI in Content Creation

AI is transforming media production:

- AI can generate:
 - Videos
 - Music
 - Images
 - Scripts

- Social media posts

Examples:

- AI-generated songs and beats
- AI editing tools for creators
- AI thumbnail generation

Companies like OpenAI and Google are leading AI media tools.

7. AI Predicting Content Performance

AI can analyze:

- Audience behavior
- Watch time
- Engagement patterns

Platforms use AI to:

- Predict views
- Recommend content
- Optimize posting times

This is why **short-form content (Reels/TikTok)** becomes addictive. AI constantly learns what keeps users watching.

8. AI in Music Creation

AI tools can:

- Generate beats
- Clone voices
- Produce full songs

Example discussion around AI music has involved tech leaders like Jeff Bezos because AI is heavily tied to cloud computing power.

AI workloads often run on **cloud servers**, especially for audio and video generation.

9. AI in Books & Copyright

AI can:

- Help write books
- Generate outlines
- Edit grammar and structure

Important:

- AI-assisted books **can be copyrighted**, but:
 - The human must contribute creative input.

AI in Everyday Technology

10. Facial Recognition & Tracking

Modern smartphones use AI facial tracking for:

- Face unlock
- Video filters
- Eye tracking in video calls

Example:

- FaceTime uses AI to keep faces centered during calls.

Developed by Apple Inc..

11. AI in Job Applications

AI is used to:

- Scan resumes (ATS systems)
- Match skills to job roles
- Recommend candidates

This is why:

- Keywords in resumes matter.

Global Trends & Social Impact

12. Aging Population (China Example)

China currently has:

- More older adults than young people (due to past population policies).

This increases demand for:

- Automation
- AI healthcare
- Smart infrastructure

13. AI Workloads (Key Concept)

AI workloads = the computing power needed to run AI models.

Examples:

- Music generation
- Video editing
- Traffic prediction
- Facial recognition

Most workloads run in:

- Cloud computing

- Data centers
- Edge devices